

Jason Tyler Carlson
Independent Researcher
New York, NY
www.jasoncarlson.org
917-439-0051
jcarlson212@gmail.com
github.com/jcarlson212

Education

Arizona State University - Master of Arts, June 2021

- Major: Mathematics

Arizona State University - Bachelor of Science, June 2020

- Major: Mathematics
- Second Major: Computer Science
- Outstanding Junior Award: Math Department

Publications

Conference Papers

- Carlson, J. T., Bernard, L., Carlson, B. H. (2026). *Beyond Imitation: Preference-Optimized Policies for Realistic Grandmaster Chess*. Proceedings of the IEEE Conference on Games (CoG 2026). Accepted for Oral Presentation. [Link](#).
- Carlson, J. T. and Carlson, B. H. (2025). *The Use of AutoML for Predicting Intracranial Aneurysm Rupture*. Congress of Neurological Surgeons (CNS 2025). Poster. [Link](#).

Industry Publications

- Moura, J., Carlson, J. T., Singh, J., Xi, S., Koparkar, S., Sun, H., Zhang, W., & Zhang, Z. (2022). *How Amazon Search Reduced ML Inference Costs by 85% with AWS Inferentia*. AWS Machine Learning Blog.
<https://aws.amazon.com/blogs/machine-learning/how-amazon-search-reduced-ml-inference-costs-by-85-with-aws-inferentia/>

Working Papers

- Carlson, J. T. (2026). *Matilda: Engine-Agnostic Search and Human-like Attention for Chess*. arXiv:2606.25176. Manuscript in preparation for AAAI. <https://arxiv.org/abs/2606.25176>
- Carlson, J. T. (2026). *TopoExplore: Topological Discrimination for Archive-Based Exploration*. arXiv:2607.09971 (preliminary report; expanded evaluation, theory, and algorithm in preparation for ICLR). <https://arxiv.org/abs/2607.09971>

Fellowships and Awards

- Outstanding Junior Award, Department of Mathematics, Arizona State University, 2020.
- Dean's Scholarship, Arizona State University..

Professional Experience

Amazon

Software Engineer II

May 2025–April 2026

- Led development of machine learning systems for credit extension and risk modeling.
- Built large-scale inference services and ML infrastructure supporting production models.
- Worked on reinforcement learning, foundation model fine-tuning, and distributed ML systems.

AppLovin

Senior Software Engineer

October 2024–March 2025

- Developed large-scale identity graph infrastructure and distributed data systems.
- Reduced infrastructure costs and improved latency for production ML pipelines.

Amazon

Software Engineer I to II

June 2021–September 2024

- Built production ML data pipelines and large-scale embedding infrastructure.

Selected Conference and Invited Presentations

- "Beyond Imitation: Preference-Optimized Policies for Realistic Grandmaster Chess," IEEE Conference on Games (CoG), 2026. (*Oral Presentation*).
- "The Use of AutoML for Predicting Intracranial Aneurysm Rupture," Congress of Neurological Surgeons (CNS), 2025. (*Poster*).

Technical Skills

Languages: Python, C++, Java, SQL

ML: PyTorch, JAX, CUDA, distributed training, RLHF/DPO-style preference optimization

RL & Simulation: Gymnasium, MiniGrid, habitat-sim

Topology & Math: GUDHI, persistent homology, algebraic topology, convex optimization

Data & Infrastructure: Apache Spark, Ray, AWS (EC2/ECS/SageMaker/Inferentia), Google Cloud

Selected Coursework

Real Analysis I (A), Real Analysis II (A+), Topology I (A+), Topology II (A), Intro to Artificial Intelligence (A), Data Structures & Algorithms (A+), Functional Analysis II (A), Abstract Algebra II (A).

Research Grants

- None to date.

Open Source

- *GarryChess*—a project for making chess engines play chess more realistically at the high Elo (Grandmaster) band. The newest model, Matilda, is planned to be released end of July. [Project Website](#). [GitHub Organization](#).
- *TopoGym*—gym environments for reinforcement learning that help train / benchmark topological navigation skills. <https://github.com/jcarlson212/TopoGym>